

Cubas-Juhn-Silos

Overview

This document describes the attached codes and data that replicate the tables and figures in the paper “Coordinated Work Schedules and the Gender Wage Gap”, by G. Cubas, C. Juhn and P. Silos

Statement about Rights

We certify that we have legitimate access to and permission to use the data used in this manuscript. We certify we have documented permission to redistribute/publish the data contained within this replication package.

The American Time Use Survey (ATUS) (Bureau of Labor Statistics (2014)) are in the public domain. The ONET data (O*NET OnLine (2018)) are licensed under a Creative Commons Attribution 4.0 International License (<https://creativecommons.org/licenses/by/4.0/>). The American Community Survey (ACS) data for 2018 and a few extra variables (see below) for the ATUS were obtained from IPUMS (Steven Ruggles, Sarah Flood, Ronald Goeken, Megan Schouweiler and Matthew Sobek (2022) and Sarah M. Flood and Backman (2012)). IPUMS does not allow for redistribution. We have obtained permission to share these data for replication of this publication only. Anyone analyzing the data for other purposes must submit a separate extract request directly via IPUMS USA or IPUMS ATUS. You may not redistribute these data without permission. Contact ipums@umn.edu for all redistribution requests. For the Cross National Equivalent Files data (Cross-National Equivalent File (2012)) we only obtained permission to share derived/analysis data. Due to redistribution restrictions we do not provide the data for the 2004 Current Population Survey 2004 supplement (Current Population Survey, May 2004: Work Schedules and Work at Home Supplement (2020)).

Data Sources and Availability

The ATUS data were obtained from the Bureau of Labor Statistics. The data can be downloaded from <https://www.bls.gov/tus/data.htm>. We use the atus-cps multi-year files from 2003 through 2014. A copy of the data is provided as part of this archive.

The ONET data are taken from two different releases, the 24.2 release (available at https://www.onetcenter.org/dl_files/database/db_24_2_excel.zip) and the 21.3 version (available at https://www.onetcenter.org/dl_files/database/db_21_3_excel.zip).

The American Community Survey (ACS) data for 2018 and the variables *WT06*, *occ*, *age*, *Uhrswrk1* for the ATUS were obtained from IPUMS. They can be downloaded after registering and filling the required documents to obtain authorization. IPUMS extended us a permission to provide a copy of the data as part of this archive for replication purposes.

Some of the data used in the appendix cannot be made publicly available. In Section A.3 of the appendix we used data from the Cross National Equivalent Files. Due to restrictions on the redistribution of the data we have obtained permission to share derived/analysis data for

replication purposes. Data can be downloaded from <https://www.cnefdata.org/data/cnef-data>. Choose <https://www.cnefdata.org/documentation/partner-policies> and follow the instructions to fill out the registration form for the USA PSID-CNEF Data Files. To access the PSID-CNEF files, users must first register with the PSID at the University of Michigan to obtain a username and password and agree to its Conditions of Use. We use the p-equiv files from 1970 through 2007.

In Section A.4 of the appendix we use the Work Schedules and Work at Home supplement of the Current Population Survey of May, 2004. We obtained the data from the Inter-University Consortium for Political and Social Research (<https://www.icpsr.umich.edu/web/ICPSR/studies/4346>). Data is subject to a redistribution restriction but can be freely downloaded from the web site if the institution of the user is part of the consortium (after registering as a user). We do not provide a copy of the data.

Software and Execution Time

The entire code ran in a Dell Inspiron 15 3000 laptop running Debian 11 Linux, Stata 17 for Linux and R version 4.1.2 in slightly less than 9 hours.

Requisites:

- A $\text{\LaTeX}$ compiler.
- Stata for Linux.
- R for Linux: the code uses the following libraries: `ggplot2`, `Hmisc`, `pracma`, `nloptr`, `data.table`, `doBy`, `minpack.lm`, and `batch`. Installing these libraries can be done by writing `install.packages("library name")` within R.
- A minimum of 5GB of RAM.

Running the Code

Extract the .zip file and enter in the folder `Codes_CubasJuhnSilos`. Besides the different folders and this readme file, there is only one file with code :

```
run_EJ_replicate_CubasJuhnSilos.sh
```

This bash script is the *only code* that needs to be run in order to generate the results. In a Linux or Mac terminal type:

```
time ./run_EJ_replicate_CubasJuhnSilos.sh
```

All the necessary code runs with that command. The script will move across folders respecting the structure of the folder tree. No additional commands or changes are needed from the user. In addition to that file, there are four folders:

- `Raw_Data`

- EJ_replicate_atus
- EJ_replicate
- EJ_replicate_model
- Penalty_Appendix

Raw Data

The data needed to run the code is all located in Raw_Data (and its subfolders). There are three main datasets we use in the paper: the American Time Use Survey (ATUS), the O*NET, and the American Community Survey (ACS)

The Raw_Data folder has five subfolders: ONET, Get_ACS_Raw_Data, Get_Atus_Raw_Data, psid, and occ_labels. The last folder — occ_labels — only has code to get into Stata format the Standard Occupational Classification codes for the different occupations we use. There is also a text file occ_labels_22 with the occupational classification we use in the model section of the paper. A subsection of the Appendix also uses the CNEF-PSID data (described in the paper) and the folder psid has the data necessary to perform the analysis.

ATUS Data

The folder Get_ATUS_Raw_Data has the following Stata do files, which the bash script executes sequentially.

- atusact_raw_0314.do (activity file)
- atusresp_raw_0314.do (respondent file)
- atusrost_raw_0314.do (roster file)
- atuscps_raw_0314.do (cps variables)
- atus_weights.do (atus weights)
- get_weights.do (atus weights)

These files get data from the original .dat files (which were downloaded from the Bureau of Labor Statistics (BLS)) and transform them into .dta (stata format) data files. All these .dta files are placed into the Stata_Raw_Data folder created in line 4 of the script.

Executing the .do files in Get_Atus_Raw_Data places the following .dta files in the Stata_Raw_Data folder:

- 1_atusact_raw_0314.dta
- atuscps_raw_0314.dta
- atusresp_raw_0314.dta
- atusrost_raw_0314.dta

- `weight_atus.dta`
- `wt_ATUS_2011_2014_new.dta`

The paper also uses a 2004 shiftwork supplement from the ATUS. Data from this supplement can't be distributed (see above) but we still include the .do file `setup_shiftwork_raw_data.do` that takes raw data and places the file `shifwrk_raw.dta` in the `Stata_Raw_Data` folder. The two files that need to be downloaded from ICPSR are `0436-0001-Data.txt` and `0436-0001-Setup.dct`.

ACS Data

Executing the .do file `get_acs.do` in the folder `Get_ACS_Raw_Data` generates the file `ACS2018.dta`, placed in the `Stata_Raw_Data` folder.

O*NET Data

Finally, the O*NET data is in the folder `Raw_Data/ONET`. An excel spreadsheet `10_18_crosswalk.xls` has the crosswalks in an easier-to-use format but taken from the Census at

<https://www2.census.gov/programs-surveys/demo/guidance/industry-occupation/2018-occupation-code-list-and-crosswalk.xlsx>

The ONET data are taken from two different releases, the 24.2 release (available at https://www.onetcenter.org/dl_files/database/db_24_2_excel.zip) and the 21.3 version (available at https://www.onetcenter.org/dl_files/database/db_21_3_excel.zip) The data in subfolders `Basic_5`, `additional_Onet_characteristics`, and `nonroutine_and_routine_onet` are from the 24.2 release and the data in the subfolder `8_onet_characteristics` come from the 21.3 release.

The .do files `do_additional_onet.do`, `do_10.do`, and `dofile_rout.do` take data from the spreadsheets and puts it into more convenient .dta files. The following .dta files are placed in the `Stata_Raw_Data` folder:

- `onet_merged_char_10.dta`
- `onet_merged_char_additional.dta`
- `merged_char_rout.dta`

In the folder labeled `psid`, we include `cpius.csv`, which contains annual data from 1970 to 2009 of the US Consumer Price Index available at <https://fred.stlouisfed.org/series/CPIAUCSL>. It also includes code to link the variables in the data files provided by CNEF, `dat_psid.R`. As explained above, the original data files are not included, but we include that code for completeness. Another R script `clean_psid_penalty.R` takes the linked dataset and creates the datafile `clean_psid_data.csv`. As explained below, this dataset is the one used to generate Table A.3 and Figure 7 in the appendix.

Once these steps are completed, all the data needed in subsequent Stata codes are in the folder `Stata_Data_Raw`. The codes will use intermediate `.dta` files created as the different `.do` files are executed, but these intermediate `.dta` files are created by the raw `.dta` files in `Stata_Raw_Data`.

Constructing Activity-Time Matrices from ATUS

The folder `EJ_replicate_atus` constructs the activity-time matrices.

- `1_matrix.do`: cleans the raw ATUS activity data and saves it as `2_matrix.dta`. As an intermediate step this code saves 134772 `.dta` individual files that are erased after `2_matrix.dta` is saved. **This is the most time-consuming step in the entire replication package; it takes several hours to run.**
- `3_matrix.do`: creates the 24X7 matrices and saves the data as `5_wrk_hcare_id.dta` (`3_matrix_merged.dta` is also saved as an intermediate step)
- `5_matrix.do`: keeps the year, id and tucaseid variables from `2_matrix` data. Saves data as `6_year_tucaseid.dta`.
- `6_matrix.do`: this is the final do file that merges the tucaseid, id and year variables with the `5_wrk_hcare_id.dta` file and the final data is saved as `1.2_data_#1.dta`. This data set contains Household variable (tucaseid), Person line number (tulineno, this is =1 since these are all ATUS individuals), for each person, time spent in each 24 time bin on work and household care, the actual time bins (goes from 1 to 24 for each individual), the exact start and stop time (start_time and stop_time), year, unique individuals (I.d).

Constructing *8to5ratio* and Empirical Evidence

In the folder `EJ_replicate/document_data_new/document_dta`, the following Stata do files merge the raw ATUS files:

- `2_1_atus_extracts_raw_0314.do`
- `2_2_atus_data_1_merge.do`
- `2_3_atus_data_1_occs.do`
- `2_4_data_2.do`

The execution of these do files generates the following stata `.dta` files (placed in the same folder).

- `2.1_atus_data#1_merge.dta`
- `2.2_2_atus_data_1_merge.dta`
- `2.3_atus_data#1_occs.dta`
- `2.4_data#2.dta`

The construction of concentration ratios and *ratio8to5* (for different levels of aggregation of occupational groups) and their merging with the main ATUS file `2.4_data#2.dta` is done in the subfolder `3_br_cr`. The following do files construct the bunching measures (*ratio8to5* and concentration ratios):

- `3a_br94.do`
- `3b_br563.do`
- `3c_cr94.do`
- `3d_cr563.do`
- `3i_br22.do`
- `3j_br22.do`

The following .dta files are created in the process:

- `3b_br563.dta`
- `3c_cr94.dta`
- `3d_cr563.dta`
- `3i_br22.dta`
- `3j_cr22.dta`
- `3_data_#3.dta`
- `3a_br94.dta`
- `3b_br563.dta`
- `3c_cr94.dta`

To merge the spousal information `4_spouses.do` creates two intermediate .dta files — `file1.dta` and `file2.dta` — and one final file `4.spouses.dta`. The latter stata data file is merged with the CPS data and the bunching ratios calculated in the `3_br_cr` subfolder in `5_data_5.do`. The resulting dataset is `5_data_#5.dta`

The data to estimate the main regressions in the paper is constructed in the folder `6_regression` data and its subfolders. There are four additional subfolders that extract variables that are included in the analysis (e.g. measures of overwork, average levels of education, etc.)

- `6_a.do`
- `/6.1_shiftwork/shift563.do`
- `/6.1_shiftwork/shift94.do`
- `/6.1_shiftwork/shift22.do`
- `/6.2_ratios/ratios_94_563_22.do`
- `/6.4_avgeduc/6.4_avg_educ.do`

- 6.5_male_overwork/6.5_male_overwork.do
- 6_b_reg.do

The final do file 6_b_reg.do creates the main regression dataset 6_b_reg.dta.

Empirical Analysis: Figures and Tables

To generate Tables 1-4, and 6-7, the script first executes 3.tab1-2.do in the folder /EJ_replicate/table1-2. The files table1.tex and table2.tex are placed in that same folder.

To replicate table 3, we need to first summarize housework categories. This is done in folder /EJ_replicate/document_data_new/hcare_detail. The execution of the following .do files generates a final dataset f6_hcaredetail_matrix_0314.dta:

- f1.do
- f2.do
- f3_a.do
- f3_b.do
- f3_c.do
- f4.do
- f5_matrix.do
- f6_hcaredetail_matrix_0316.do

There are several .dta intermediate files generated and used in the analysis of creating the final dataset. They are placed in that same folder. The script

/EJ_replicate/table3/4.tab3_hhcare_detail_reg2.do

generates table3reg.tex (this is table 3 in the paper).

Generating tables 4,6, and 7 requires some analysis of the O*NET data.

In folder /EJ_replicate/tables6-7 the following .do files merge the O*NET measures with the SOC occupational codes and perform the regressions shown on Tables 6 and 7 of the paper:

- regs_ONET_563.do
- 6.tab6.do
- 7.tab7.do

The files table6.tex and table7.tex are placed in the same folder.

The .do files that perform the correlation analysis shown in table 4 are in the folder /EJ_replicate/table4:

- generalizing_soc.do

- `main_code_to_2002.do`
- `5.tab4.do`

Running this code generates the following `.dta` files:

- `generalizing_soc.dta`
- `main_result_to_2002.dta`
- `bratio_563all.dta`
- `cratio_563all.dta`
- `ONET_563b.dta`

as well as the file `table4_corr_vals.smcl` that has the numbers shown in Table 4 in the paper. An R script below takes the information in this file to populate a $\text{\LaTeX}$ table.

To get the figures 1 - 4 , the bash script executes the following two `.do` files located in folder `/EJ_replicate/fig1-4`

- `1.fig1-3.do`
- `2.fig4_563.do`

The files `fig1.eps`, `fig2.eps`, `fig3.eps` and `fig4.eps` are saved in the same folder.

The folder `/EJ_replicate/appendix` generates the tables related to the empirical analysis and shown in the online appendix.

1. Tables A2 and A4: They are contained in the files `tableA2.tex` and `tableA4.tex`. They are constructed by the authors, not generated by any code. Tables A5 and A6: They are contained in the files `tableA5.tex` and `tableA6.tex`. They are generated by running `tabA5-A6.do`. It uses the file `5_data_#5.dta` already generated (see page 4).
2. Table A7: It is contained in the file `tableA7.tex`. It is generated by running `tabA7.do` and uses the files `main_result_to_2002.dta` (generated by `main_code_to_2002.do`, see page 5). It generates the data files `1_t3_t4_563.dta` and `1_t3_t4_563_format.dta` which are then used by `tabA7.do` to generate `tableA7.tex`.
3. Table A8: It is contained in the file `tableA8.tex`. This table is the one showing estimated regressions including the fraction of workers reporting working in shifts. The data on shiftwork, as explained at the beginning of this document, is available through the ICPSR registration system and not available. Once those data are obtained, we would run the code from the beginning including the `.do` file in the folder `Raw_data/Get_Atus_Raw_data/Shiftwork_Supplement`. After that, we would also run `tabA8.do` in the `EJ_replicate/appendix` folder. With the data included in this replication package the code `tabA8.do` will not run.
4. Table A9: It is contained in the file `tableA9.tex`. It is generated by running `tabA9.do` and uses the files `6_b_reg.dta` (generated by `6_b_reg.do`, see page 5), `ONET_563b.dta` (generated by running `regs_ONET_563.do` included in the folder `EJ_replicate/tables6-7`,

see page 6) and `cratio_563all.dta` (generated by running `5.tab4.do` included in the folder `EJ replicate/table4`, see page 6).

Structural Model

The folder `/EJ_replicate_model` has the following three subfolders:

1. `Sect_4_Simple_Case_code`
2. `Quantitative_Analysis/model_files`
3. `Quantitative_Analysis/compute_emp_moments`

The remainder of the bash script will:

1. Create the folders `Quantitative_Analysis/data`, `Quantitative_Analysis/model_output`, `Quantitative_Analysis/sensitivity_cobb_douglas`
2. Run the code to calculate the results in Section 4. This code is in folder `Sect_4_Simple_Case_code`. The results for the three simple economies (no gender, gender through different preferences, and tastes) will be in that folder too after running the code. The two R functions that generate the results are

`/Sect_4_Simple_Case_code/simple_cases.R`,

and

`/Sect_4_Simple_Case_code/simple_cases.fun.R`. The files with the results are:

- `results_simple_case_no_gender.txt`
- `results_simple_case_gender_diff_nu.txt`
- `results_simple_case_tastes.txt`

All of these results will be used by code to generate $\text{\LaTeX}$ tables described below. A separate subfolder in `Sect_4_Simple_Case_code` called `two-earner-households`, has the code that generates the results in Section A.2 of the appendix. The R scripts are `main_code_ge.R`, `new_funs.R`, and `new_one.R`. They generate three files to be used later to generate the paper's tables

- `results_two_earner_positive.txt`
- `results_two_earner_negative.txt`
- `results_two_earner.txt`.

3. Using data in the `Quantitative_Analysis/data` folder and R code in `Quantitative_Analysis/compute_emp_moments`, a bunch of files needed for the model calibration will be placed in the folder `Quantitative_Analysis/data`. The files run are:

- `Quantitative_Analysis/compute_emp_moments/moments_byoccs_def.R`
- `Quantitative_Analysis/compute_emp_moments/decomp_gender_gap.R`

- `Quantitative_Analysis/compute_emp_moments/compute_br_22occs.R`
 - `Quantitative_Analysis/compute_emp_moments/reg_data_22occs.R`
4. The folder `Quantitative_Analysis/model_files` has the main code to generate the quantitative results from the model. There are three files `main_ge.R`, `new_one.R` and `new_funs.R`. `main_ge.R` is the main code and the two other files provide auxiliary functions/utilities. The bash script runs `main_ge.R` for each experiment changing the elasticity of substitution across occupations between the baseline case and the cobb-douglas case. The output files with model moments etc. are put into the folder `model_output`.
 5. Before generating the final tables and figures, and compiling the $\text{\LaTeX}$ file, two additional R scripts generate the appendix results in section A.3: Tables A.2 and A.3 and Figure 7. Running `penalty_regression_final.R` and `graph_table.R` generate `table_A3` and `psid_penalty.pdf`. Table A.2 is included in the file `table_A2.tex` located in the `Raw_data/psid` folder, since that table describes a mapping between occupations that the authors generated.
 6. The code `get_numbers_table_4.R`, `fix_tables.R`, `generate_tables.R` and `generate_figures.R` in the `EJ_replicate_mode/latex_tables` folder, takes all output from the previously described code and populates latex tables (tables 1-12 in the main body of the paper, and tables A.2, A.3, A.10, A.11, A.12, and A.13) and construct figures 1-7. These tables and figures are shown in the file `final_model_tables.pdf`

References

- Bureau of Labor Statistics (2014): “American Time Use Survey,” data retrieved from Bureau of Labor Statistics, <https://www.bls.gov/tus/data.htm>.
- Cross-National Equivalent File (2012): “Cross-National Equivalent File - PSID,” <https://www.cnefdata.org/about/about-cnef>.
- Current Population Survey, May 2004: Work Schedules and Work at Home Supplement (2020): United States. Bureau of the Census, and United States Department of Labor. Bureau of Labor Statistics. Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/ICPSR04346.v2>.
- O*NET OnLine (2018): “National Center for O*NET Development,” data retrieved from www.onetonline.org/help/online/data.
- Sarah M. Flood, L. C. S., and D. Backman (2012): IPUMS ATUS. American Time Use Survey Data Extract Builder: Version 3.1 [dataset]. College Park, MD: University of Maryland and Minneapolis, MN: IPUMS, 2022 <https://doi.org/10.18128/D060.V3.1>.
- Steven Ruggles, Sarah Flood, Ronald Goeken, Megan Schouweiler and Matthew Sobek (2022): IPUMS USA: Version 12.0 [dataset]. Minneapolis, MN: IPUMS, 2022. <https://doi.org/10.18128/D010.V12.0>.